Anton Xue

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Research Interests

Diffusion language models, AI for mathematics, AI safety and interpretability, formal methods.

Research Experience

The University of Texas at Austin — 2025 – Present
Postdoctoral Fellow, Institute for Foundations of Machine Learning
Advisors: Sanjay Shakkottai and Adam Klivans
Research: Diffusion language models; AI agents for mathematical research

SRI International — 2022
Research Intern
Mentor: Susmit Jha
Research: Control and verification of deep equilibrium models

Nokia Bell Labs — 2019
Research Intern
Mentor: Kedar Namjoshi
Research: Certified compilers for WebAssembly

Harvard University School of Engineering and Applied Sciences — 2018
Research Intern
Mentors: Stephen Chong and Marco Gaboardi
Research: Program synthesis and formal semantics for R

Max Planck Institute for Software Systems — 2017
Research Intern
Mentors: Rupak Majumdar and Damien Zufferey
Research: Formal verification of embedded systems

Education

University of Pennsylvania — 2019 – 2025
Ph.D. in Computer and Information Science
Advisors: Rajeev Alur and Eric Wong
Thesis: Trustworthy Machine Learning: Specification, Verification, and Explanation

Yale University — 2015 – 2019
B.S. in Computer Science and Mathematics
Advisor: Ruzica Piskac
Research: Program analysis and symbolic execution for Haskell

Publications

* denotes equal contribution.

Preprints and Technical Reports

[1] Quentin Fruytier, **Anton Xue**, Sanjay Shakkottai, and Sujay Sanghavi. PIG: Parallel Inconsistency Guidance for Accelerating Diffusion Language Models. *arXiv preprint arXiv:2610.17375*, 2026.

[2] **Anton Xue**, Alan Li, and Rahul Saha. Multi-Target Few-Shot Prompting with Diffusion Language Models. *arXiv preprint arXiv:2610.22294*, 2026.

[3] Lyuxin David Zhang, Eric Wong, Surbhi Goel, and **Anton Xue**. LESSER: Post-Training Data Selection with Output-Layer Gradients. *arXiv preprint arXiv:2610.03702*, 2026.

[4] **Anton Xue**, Litu Rout, Aditya Akella, Adam Klivans, Sujay Sanghavi, and Sanjay Shakkottai. dQwen3.5: Hybrid-Attention Diffusion Language Models. *arXiv preprint arXiv:2609.20751*, 2026.

[5] Alan Li*, Rahul Saha*, **Anton Xue**, Swarat Chaudhuri, Adam Klivans, Pravesh Kothari, and Raghu Meka. Long-Horizon AI Research for Grothendieck Constant: A Case Study in Human-AI Mathematical Collaboration. *arXiv preprint arXiv:2608.11195*, 2026.

[6] Rahul Saha*, Alan Li*, **Anton Xue**, Swarat Chaudhuri, Adam Klivans, Pravesh Kothari, and Raghu Meka. New Lower and Upper Bounds for the Grothendieck Constant. *arXiv preprint arXiv:2608.11158*, 2026.

[7] Alan Li*, Rahul Saha*, **Anton Xue**, Swarat Chaudhuri, Adam Klivans, Pravesh Kothari, and Raghu Meka. The Grothendieck Constant is Less Than $\pi/(2\log(1+\sqrt{2})) - 10^{-5}$. *arXiv preprint arXiv:2606.03991*, 2026.

[8] **Anton Xue***, Litu Rout*, Constantine Caramanis, and Sanjay Shakkottai. AnCoder: Anchored Code Generation via Discrete Diffusion Models. *arXiv preprint arXiv:2602.17688*, 2026.

[9] Shreya Havaldar*, Weiqiu You*, Chaehyeon Kim, **Anton Xue**, Helen Jin, Marco Gatti, Bhuvnesh Jain, Helen Qu, Amin Madani, Daniel Hashimoto, Gary Weissman, Rajat Deo, Sameed Khatana, Lyle Ungar, and Eric Wong. T-FIX: Text-Based Explanations with Features Interpretable to eXperts. *arXiv preprint arXiv:2511.04070*, 2025.

Peer-Reviewed Publications

[1] Rahul Saha*, Alan Li*, **Anton Xue**, Adam Klivans, Pravesh Kothari, Raghu Meka, and Swarat Chaudhuri. Optimizing Analytic Constants via AI-Guided Lean Proof Refinement. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.

[2] Shailesh Sridhar, **Anton Xue**, and Eric Wong. Missingness Bias Calibration in Feature Attribution Explanations. *International Conference on Learning Representations (ICLR)*, 2026.

[3] Andy Yang, Christopher Watson, **Anton Xue**, Satwik Bhattamishra, Jose Llarena, William Merrill, Emile Dos Santos Ferreira, Anej Svete, and David Chiang. The Transformer Cookbook. *Transactions on Machine Learning Research (TMLR)*, 2026.

[4] Helen Jin*, **Anton Xue***, Weiqiu You, Surbhi Goel, and Eric Wong. Probabilistic Stability Guarantees for Feature Attributions. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[5] Weiqiu You, **Anton Xue**, Shreya Havaldar, Delip Rao, Helen Jin, Chris Callison-Burch, and Eric Wong. Probabilistic Soundness Guarantees in LLM Reasoning Chains. *Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

[6] Helen Jin*, Shreya Havaldar*, Chaehyeon Kim*, **Anton Xue***, Weiqiu You*, Helen Qu, Marco Gatti, Daniel Hashimoto, Bhuvnesh Jain, Amin Madani, Masao Sako, Lyle Ungar, and Eric Wong. The FIX Benchmark: Extracting Features Interpretable to eXperts. *Journal of Data-centric Machine Learning Research (DMLR)*, 2025.

[7] Thomas Zhang*, Behrad Moniri*, Ansh Nagwekar, Faraz Rahman, **Anton Xue**, Hamed Hassani, and Nikolai Matni. On The Concurrence of Layer-wise Preconditioning Methods and Provable Feature Learning. *International Conference on Machine Learning (ICML)*, 2025.

[8] **Anton Xue***, Avishree Khare*, Rajeev Alur, Surbhi Goel, and Eric Wong. Logicbreaks: A Framework for Understanding Subversion of Rule-based Inference. *International Conference on Learning Representations (ICLR)*, 2025.

[9] Xiayan Ji*, **Anton Xue***, Eric Wong, Oleg Sokolsky, and Insup Lee. AR-Pro: Counterfactual Explanations for Anomaly Repair with Formal Properties. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[10] Chang Zhu, Ziyang Li, **Anton Xue**, Ati Priya Bajaj, William Gibbs, Yibo Liu, Rajeev Alur, Tiffany Bao, Hanjun Dai, Adam Doupé, Mayur Naik, Yan Shoshitaishvili, Ruoyu Wang, and Aravind Machiry. TYGR: Type Inference on Stripped Binaries using Graph Neural Networks. *USENIX Security*, 2024.

[11] **Anton Xue**, Lars Lindemann, and Rajeev Alur. Chordal sparsity for SDP-based neural network verification. *Automatica*, 2024.

[12] **Anton Xue**, Rajeev Alur, and Eric Wong. Stability Guarantees for Feature Attributions with Multiplicative Smoothing. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[13] **Anton Xue**, Lars Lindemann, Alexander Robey, Hamed Hassani, George Pappas, and Rajeev Alur. Chordal Sparsity for Lipschitz Constant Estimation of Deep Neural Networks. *Conference on Decision and Control (CDC)*, 2022.

[14] Osbert Bastani, Shuo Li, and **Anton Xue**. Safe Reinforcement Learning via Statistical Model Predictive Shielding. *Robotics: Science and Systems (RSS)*, 2021.

[15] Rajeev Alur, Phillip Hilliard, Zachary Ives, Konstantinos Kallas, Konstantinos Mamouras, Filip Niksic, Caleb Stanford, Val Tannen, and **Anton Xue**. Synchronization Schemas. *Symposium on Principles of Database Systems (PODS)*, 2021.

[16] **Anton Xue** and Nikolai Matni. Data-Driven System Level Synthesis. *Learning for Dynamics and Control (L4DC)*, 2021.

[17] Kedar Namjoshi and **Anton Xue**. A Self-certifying Compilation Framework for WebAssembly. *Verification, Model Checking, and Abstract Interpretation (VMCAI)*, 2021.

[18] William Hallahan, **Anton Xue**, and Ruzica Piskac. G2Q: Haskell constraint solving. *International Symposium on Haskell*, 2019.

[19] William Hallahan, **Anton Xue**, Maxwell Troy Bland, Ranjit Jhala, and Ruzica Piskac. Lazy counterfactual symbolic execution. *Programming Language Design and Implementation (PLDI)*, 2019.

Mentoring Experience

Quentin Fruytier (UT Ph.D. student): Faster inference with diffusion language models — 2026 – Present

Alan Li (UT Ph.D. student): AI agents for mathematical research — 2025 – Present

Rahul Saha (UT Ph.D. student): AI agents for mathematical research — 2025 – Present

David Zhang (Penn undergraduate): Post-training data selection for language models — 2023 – Present

Faraz Rahman (Penn undergraduate): Explainability and interpretability of diffusion models — 2023 – 2024

Jawad Ahmad (high school student): Programmer-friendly code synthesis with ChatGPT — 2023

Awards & Fellowships

Postdoctoral Fellowship, Institute for Foundations of Machine Learning at UT Austin — 2025

Recipient, AWS gift supporting trustworthy AI research at Penn (ASSET Center) — 2023

University of Pennsylvania ENIAC Fellowship — 2019

Yale Computer Science Award — 2019

National Science Foundation Graduate Research Fellowship — 2019

Yale College Freshman Summer Research Fellowship — 2016

Professional Service

Area Chair: ICLR — 2026 – Present

Reviewer: NeurIPS, ICML, ICLR, AAAI — 2024 – Present

Reviewer: AI-MATH, AI-Native Academia, SATA, SciForDL, AdvML — 2023 – Present

Reviewer: IEEE L-CSS, CDC, ACC, Automatica, L4DC — 2021 – Present

Artifact Evaluation Committees: TACAS (2023), SAS (2021), VMCAI (2021), PLDI (2020, 2021) — 2020 – 2023

Teaching Experience

University of Pennsylvania, Teaching Assistant: Convex Optimization; Fundamentals of Linear Algebra and Optimization; Mathematical Foundations of Computer Science — 2020 – 2023

Yale University, Teaching Assistant / Course Grader: Real Analysis; Intensive Algorithms; Design and Analysis of Algorithms; Mathematical Tools for Computer Science — 2016 – 2019